The Lorenz Attractor

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1 Introduction

Chaos theory is studied in physics in order to understand systems which are complex and chaotic in nature. To be chaotic, a system must obey deterministic equations of motion, yet produce future behavior that is unpredictable. These equations of motion are typically nonlinear in nature. These states of disorder are usually caused by the initial conditions of the system for the particular problem.

Chaos is significant in that it can be seen in a variety of areas, including: chemical reactions, lasers, the spread of disease, and climate and weather. This problem deals with a system of three first-order differential equations, developed by Edward Lorenz in 1963 to model atmospheric convection:

$$\dot{x} = \sigma(y - x) \tag{1}$$

$$\dot{y} = x(\rho - z) - y \tag{2}$$

$$\dot{z} = xy - z \tag{3}$$

As we will see, given certain parameters and initial conditions, these three equations will display either a chaotic or nonchaotic motion. The problem requires the following: For three different scenarios for the Lorenz system of equations, find the equilibrium points of the system, test the stability of each equilibrium point, and for the second and third scenarios, use the Liapanov Exponent as a means of deciding whether the system exhibits signs of chaos. For this problem, we are given β= 4, σ=28, and ρ will vary.

The equation which we will use to find the Liapanov Exponent is Eqn. 12.26 from the *Classical Mechanics* by Taylor: $|\Delta\phi(t)| \sim Ke^{\lambda t}$ where K is a positive constant and λ is the Liapanov Exponent. If the Liapanov Exponent is positive, then the long-term motion of the system is chaotic. If the Liapanov Exponent is negative, then the long-term motion is nonchaotic. For this problem, rather than solving this equation for λ analytically, we will solve for the system's motion numerically over 20 seconds. We will plot the time vs natural log of the distance between the points, and the slope of the line of best fit for this plot will be the Liapanov Exponent for the particular scenario.

We will find the equilibrium points for the Lorenz system of equations by

optimizing the functions the way which we have done many other times this semester: by setting each of the three first-order differential equations to zero, and solving for x, y and z. This will provide us with the general form (in terms of the parameters $x, y, z, \beta, \rho, \sigma$) of all the equilibrium points. This process is shown below:
First, set Eqn. 1 equal to 0.

$$0 = \sigma(y - x),$$

Then $\sigma = 0$ and $y = x$. Now set Eqn. 2 equal to 0.

$$0 = x(\rho - z) - y$$

$$0 = x\rho - xz - y$$

and since $y = x$

$$0 = x\rho - xz - x$$

$$0 = x(\rho - z - 1)$$

Then $x = 0$ and $z = \rho - 1$. Set Eqn. 3 equal to 0.

$$0 = xy - \beta z$$

and since $y = x$

$$0 = x^2 - \beta z$$

$$x = x^2 - \beta z$$

Then $x = \pm\sqrt{\beta z}$.
As a result, we retrieve the general format for equilibrium points for this system by plugging in x, y and z for the components of the points for the different scenarios we will now work through:

$$P_1 = (0, 0, 0) \tag{4}$$

$$P_2 = (\sqrt{\beta(\rho - 1)}, \sqrt{\beta(\rho - 1)}, \rho - 1) \tag{5}$$

$$P_3 = (-\sqrt{\beta(\rho - 1)}, -\sqrt{\beta(\rho - 1)}, \rho - 1) \tag{6}$$

2 Part A

For each scenario it is known that β=4 and σ=28, and ρ will vary. For part A, ρ is given as 0.5. Plugging these parameters into the general form of equilibrium points we found above, we find that $P_1 = (0, 0, 0)$, $P_2 = (\sqrt{2}i, \sqrt{2}i, -0.5)$, and $P_3 = (-\sqrt{2}i, -\sqrt{2}i, -0.5)$. The next step is to solve for the stability of each of these equilibrium points. This is accomplished per the problem instructions, by choosing a nearby point as the initial placement, then solving for the motion of

the system numerically over 20 seconds, and lastly plotting the resultant motion. Because the second and third equilibrium points are made up of imaginary components, we need only complete this process for (0,0,0), as shown below using Scipy's Odeint.

As we have learned this semester when solving for stability in equilibrium points, a point is considered stable when nearby points tend to move closer to the equilibrium point, and is considered unstable when nearby points tend to move further away from the chosen nearby point. The plot in Figure 1 illustrates that the equilibrium point (0,0,0) is unstable, because the motion of the system around the chosen nearby point (0.01, 0.01, 0.01) is deviating ("running away") from this chosen point. There are two lines rather than three despite there being three solutions because as we saw previously x is equal to y therefore their solution sets match.

```
def DE(state, time):
    # variables in DE
    x, y, z = state
    # Constants and initial conditions
    # where B is beta, p is rho, and o is sigma
    B = 4
    p = 0.5
    o = 28
    # Right hand side of equations
    derivs = [o*(y-x), x*(p-z)-y, x*y-B*z]
    return derivs
```

```
# choosing nearby point (0.01, 0.01, 0.01)
xi = 0.01
yi = 0.01
zi = 0.01
ts = np.arange(0, 20, 0.01)
sols = odeint(DE, [xi, yi,zi], ts)
```

3 Part B

For part B, ρ is given as 10. Plugging this along with the given parameters into the general form of equilibrium points, we find that $P_1 = (0, 0, 0)$, $P_2 = (6, 6, 9)$ and $P_3 = (-6, -6, 9)$. We will find the stability of these points by completing the same process as in Part A; generating plots of the system's motion around points nearby these equilibrium points. We see that when ρ=10, the equilibrium point (0,0,0) is unstable, because the motion of the system is "running away from" the initial point (0.01, 0.01, 0.01). In contrast, we find that the points (6,6,9) and (-6,-6,9), shown in Figures 2 and 3, are stable equilibrium points because the motion of the system around the chosen nearby points is stagnant throughout the entire time. Therefore, in comparison to Part A, the equilibrium

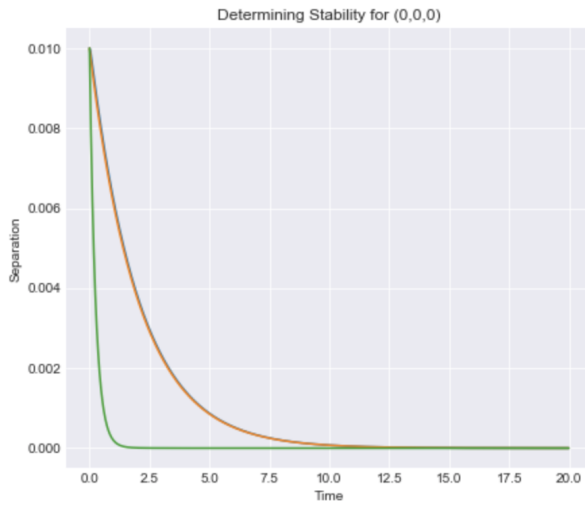

Figure 1: Using the point (0.01, 0.01, 0.01) to determine stability of equilibrium point (0, 0, 0).

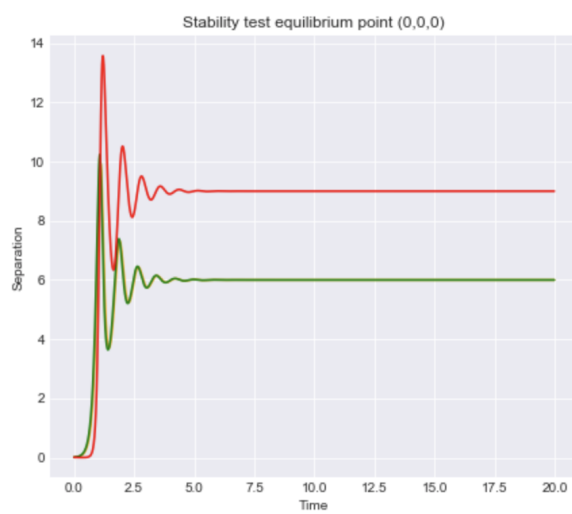

Figure 2: Using the point (0.01,0.01,0.01) to test the stability of equilibrium point (0,0,0)

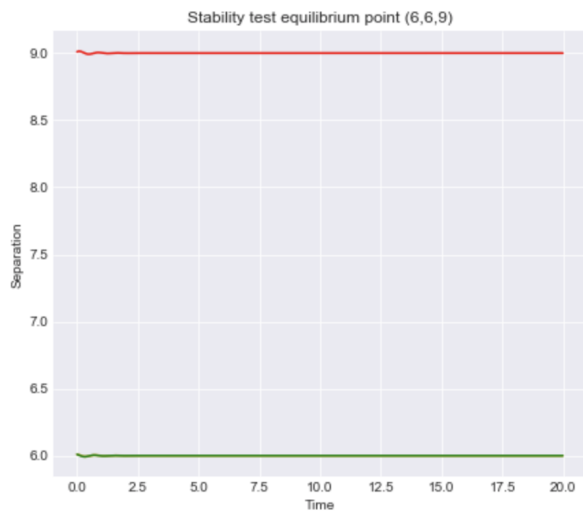

Figure 3: Using the point (6.01, 6.01, 9.01) to test the stability of equilibrium point (6,6,9)

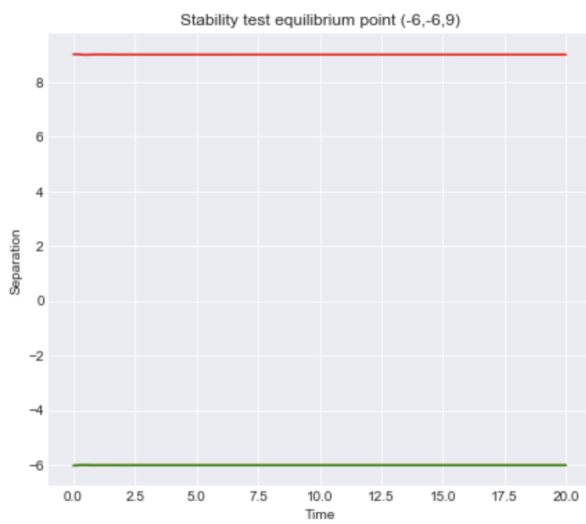

Figure 4: Using the point (-6.01, -6.01, 9.01) to test the stability of equilibrium point (6,6,9)

point (0,0,0) is still unstable, however there are two more equilibrium points as with ρ=10 we no longer have equilibrium points made up of imaginary number components.
The next step is to solve for the Liapanov Exponent. This is the trickiest part, as we are solving for the exponent numerically through Odeint. The Liapanov Exponent can be solved for numerically by choosing two points that are 0.001 distance from each other, but some distance from any of the equilibrium points. Choosing the points (1,1,1) and (1.001, 1.001, 1.001), we solve the system of equations through odeint once more. Being careful here, we recognize that this means we will have two different sets of initial conditions and therefore two different solution sets. It is then necessary to find the distance between the points. It can be tempting to solve for the difference between components of each solution set, but to solve for the distance between the points it is necessary to solve for the difference between the entire solution sets to each system of differential equations. It is then possible to plot the times versus the natural log of the distance between the points. By looking once more at equation 12.26 we see how taking the natural log of the distance between the poits solves for λ, the Liapanov Exponent. Plotting the motion in scatter plot form generates a plot

```
ts = np.arange(0, 20, 0.01)
xi = 1
yi = 1
zi = 1
vxi = 0
vyi = 0
vzi = 0

x2i = xi + .001
y2i = yi + .001
z2i = zi + .001
sols2 = odeint(DE_10, [x2i, y2i, z2i], ts)
sols = odeint(DE_10, [xi, yi, zi], ts)
```

```
difference = sols-sols2
difference
```

```
array([[-1.00000000e-03, -1.00000000e-03, -1.00000000e-03],
       [-1.00895162e-03, -1.07012333e-03, -9.81213476e-04],
       [-1.03295759e-03, -1.14133005e-03, -9.65046432e-04],
       ...,
       [-1.59516844e-11, -1.86295424e-11,  5.34861044e-12],
       [-1.59126046e-11, -1.85869098e-11,  5.33972866e-12],
       [-1.58744129e-11, -1.85442772e-11,  5.33262323e-12]])
```

which can have a line of best fit applied through a polyfit function. The slope of this line of best fit is the Liapanov Exponent. In the case of Part B with ρ=10, the slope of the line is found to be approximately -0.93. This indicates that the long term motion of the system is nonchaotic, as the Liapanov Exponent is negative. Therefore system does not exhibit signs of chaos.

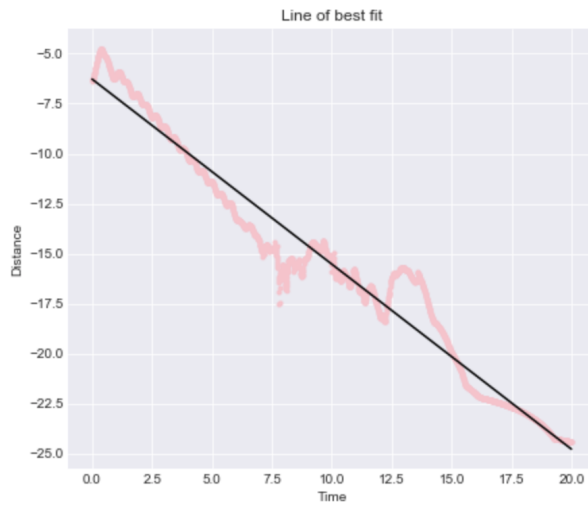

Figure 5: Generating a line of best fit over plot of times versus distance between points.

4 Part C

For part C we have the same values as before for σ and β, and $\rho = 46.92$. Plugging these in to solve for the equilibrium points, we find that $P_1 = (0, 0, 0)$, $P_2 = (13.55, 13.55, 45.92)$ and $P_3 = (-13.55, -13.55, 45.92)$. This is similar to Part B in that we have three equilibrium points as the value of ρ is large enough such that there are no longer imaginary numbers.
Next we solve for the stability of each equilibrium point by using odeint to solve the system of equations with initial values close to the equilibrium points. In this case, these chosen nearby values are (0.01,0.01,0.01), (13.56,13.56,45.93) and (-13.56,-13.56,45.93). From our plots we see that the equilibrium point (0,0,0) is still unstable as was found in Part B. This is shown in that although the oscillation pattern around 0.01 is more different than in Part B, its behavior is similar because the motion deviates toward the equilibrium points that are stable, such as (13.56,13.56,45.93). Similarly, the other two equilibrium points (13.56,13.56,45.93) and (-13.56,-13.56,45.93) are stable, which is shown in how the resultant motion is stagnant around this point which is nearby the original equilibrium point. In this way, the stability of these equilibrium points is the same as the stability found for the equilibrium points in Part B, because in both cases (0,0,0) is unstable while the other two equilibrium points are stable.

As in Part B we will again numerically solve for the Liapanov Exponent. The same points will be used in odeint to solve the system of equations twice, resulting in two sets of solutions. The distance between points is again found by taking the difference of the solution sets. The line of best fit is generated by a polyfit function and taking the slope of this line we find λ is equal to

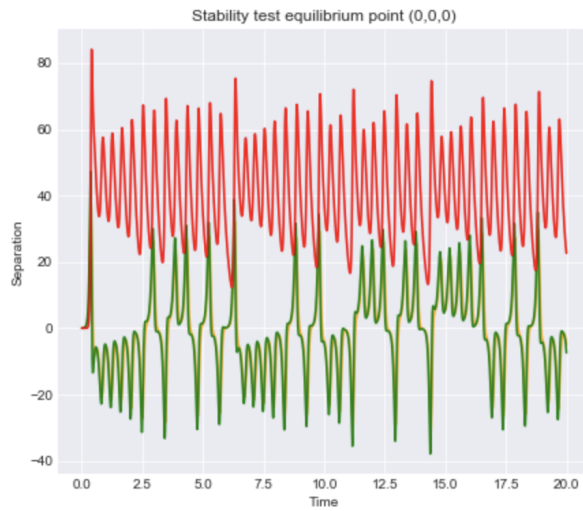

Figure 6: Using nearby point (0.01,0.01,0.01) to test the stability of equilibrium point (0,0,0). Equilibrium point is unstable.

Figure 7: Using nearby point (13.56,13.56, 45.92) to test the stability of equilibrium point (13.55, 13.55, 45.92).Equilibrium point is stable.

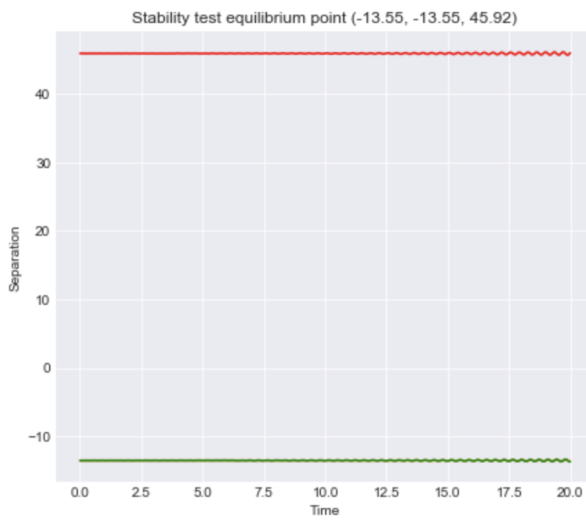

Figure 8: Using nearby point (-13.56,-13.56, 45.92) to test the stability of equilibrium point (-13.55, -13.55, 45.92). Equilibrium point is stable.

approximately 0.41. As seen in figure 9, this system exhibits signs of chaos because the Liapanov Exponent is positive, which we know from Eqn. 12.26 in the text indicates that for long-term motion the system will exhibit chaos. This is interesting because we noticed that the stability for equilibrium points in this case was the same as in Part B, therefore one might believe this is an indicator that this third scenario for the system of equations also produces nonchaotic motion over the long-term, however this is not true. This is significant because it shows that while the system obeys the deterministic laws of motion, a slight change in initial conditions can bring about chaotic behavior in the future.

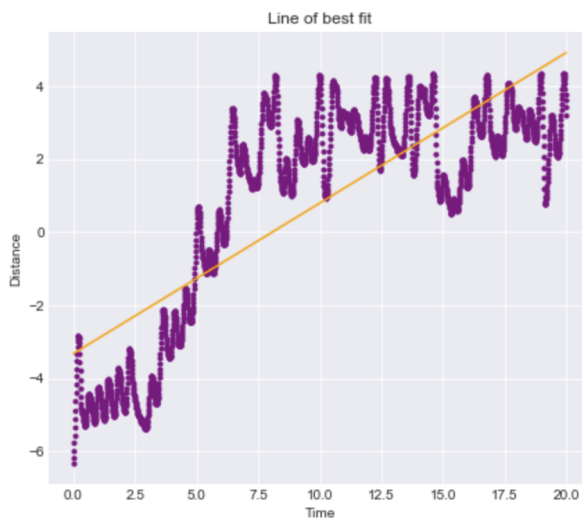

Figure 9: Generating a line of best fit over plot of times versus distance between points.